DE LA RECHERCHE À L'INDUSTRIE

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COLLISION OPERATOR IN GYSELA

GYSELA TEAM

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Three main parts of the collision operator

Step 1: Computation of the moments of the distribution function

- subroutines « collvparmu_general_moments » and « collvparmu_general_q_para_on_naTa »

Step 2: Modification of the distribution function by collisions

$$\begin{cases} \dfrac{\partial F_{M0a}}{\partial t} = \displaystyle\sum_b C^0_{ab}\left(F_{M0a}, F_{M0b}\right) \\ \dfrac{\partial F_a}{\partial t} = \displaystyle\sum_b C_{\parallel,ab}\left(F_a, F_b\right) \qquad \left(F_{M0a} = cst\right) \\ \dfrac{\partial F_a}{\partial t} = \displaystyle\sum_b C_{v,ab}\left(F_a, F_b\right) + C_{d,ab}\left(F_a, F_b\right) \qquad \left(F_{M0a} = cst\right) \end{cases}$$

$$F_{M0a}(\boldsymbol{x}, \boldsymbol{v}, t) = n_a(\boldsymbol{x}, t)\left(\frac{1}{2\pi\, v_{Ta}^2}\right)^{3/2} \exp\left(-x_a^2\right)$$

[P. Donnel 2019]

- subroutines « collvparmu_general_evolution_Maxwellian », « collvparmu_general_Cpar » + Others

Step 3: Correction procedure to improve conservation of density, momentum, energy

- subroutines « collvparmu_general_correction_density » and «collvparmu_general_correction_Vpar_T »

Cv+Cd an advection-diffusion in velocity space

$$\frac{\partial F_a}{\partial t} = \sum_b C_{v,ab}\left(F_a, F_b\right) + C_{d,ab}\left(F_a, F_b\right)$$

$$C_{v,ab}\left(F_a\right) = \frac{1}{2v_\perp}\frac{\partial}{\partial v_\perp}\left[F_{M0a}\nu_{v,ab}v_\perp^2\left(v_\perp\frac{\partial g_{ab}}{\partial v_\perp} + v_\parallel\frac{\partial g_{ab}}{\partial v_\parallel}\right)\right] + \frac{1}{2}\frac{\partial}{\partial v_\parallel}\left[F_{M0a}\nu_{v,ab}v_\parallel\left(v_\perp\frac{\partial g_{ab}}{\partial v_\perp} + v_\parallel\frac{\partial g_{ab}}{\partial v_\parallel}\right)\right]$$

$$C_{d,ab}\left(F_a\right) = \frac{1}{2v_\perp}\frac{\partial}{\partial v_\perp}\left[F_{M0a}\nu_{d,ab}v_\perp v_\parallel\left(v_\parallel\frac{\partial g_{ab}}{\partial v_\perp} - v_\perp\frac{\partial g_{ab}}{\partial v_\parallel}\right)\right] + \frac{1}{2}\frac{\partial}{\partial v_\parallel}\left[F_{M0a}\nu_{d,ab}v_\perp\left(-v_\parallel\frac{\partial g_{ab}}{\partial v_\perp} + v_\perp\frac{\partial g_{ab}}{\partial v_\parallel}\right)\right]$$

$$g_{ab} = g_a = f_a - \frac{m_a v_\parallel U_{\parallel d,a}}{T_a} \qquad f_a = \frac{F_a}{F_{M0a}} \qquad U_{\parallel d,a} = V_{\parallel a} - \frac{q_{\parallel a}}{n_a T_a}\left(1 - \frac{2}{5}x_a^2\right)$$

Treatment of $v_\perp$ derivatives

- Limited number of points in $\mu = \frac{mv_\perp^2}{2B}$

➢ Projection on a set of orthogonal polynomials (Laguerre)

$$F(\boldsymbol{r}, v_\parallel, u, t) = F_{M0} \sum_l \alpha_{\ell,a}(\boldsymbol{r}, v_\parallel, t) P_\ell(u)$$

with $u = \frac{\mu B}{T}$

Consequences :

- μ derivatives are treated analytically
- Integrals on μ direction have to be performed (projections)
- Add analytical complexity to the problem (projection of the collision operator on the polynomials)

Motivations :

- Better conservation properties than with finite differences

Cv+Cd: projection on Laguerre polynomials

$$\frac{\partial F_a}{\partial t} = \sum_b C_{v,ab}\left(F_a, F_b\right) + C_{d,ab}\left(F_a, F_b\right) \Leftrightarrow \frac{\partial f_a}{\partial t} = \sum_b \bar{C}_{ab}$$

$$\bar{C}_{ab}(g_a) = K_{1,ab}\frac{\partial g_a}{\partial v_\parallel} + K_{2,ab}\frac{\partial^2 g_a}{\partial v_\parallel^2} + K_{3,ab}u\frac{\partial^2 g_a}{\partial v_\parallel \partial u} + K_{4,ab}u\frac{\partial g_a}{\partial u} + \bar{C}_{m,ab}(g_a)$$

$$\frac{\partial \alpha_{\ell,a}}{\partial t} = \sum_j \left[\alpha'_{i,a}N^{jl}_{0,a} + \frac{\partial \alpha'_{i,a}}{\partial v_\parallel}N^{jl}_{1,a} + \frac{\partial^2 \alpha'_{i,a}}{\partial v_\parallel^2}N^{jl}_{2,a}\right]$$

$$N^{jl}_{0,a} = \sum_b \left\langle jK_{4,ab}\left(P_j - P_{j-1}\right) + \bar{C}_{m,ab}(P_j)|P_l\right\rangle$$

$$N^{jl}_{1,a} = \sum_b \left\langle K_{1,ab}P_j + jK_{3,ab}\left(P_j - P_{j-1}\right)|P_l\right\rangle$$

$$N^{jl}_{2,a} = \sum_b \left\langle K_{2,ab}P_j|P_l\right\rangle$$

Can be expressed analytically by recurrence (cf paper for more details)

Cv+Cd: steps of computation

Step 1: $F_a \longrightarrow f_a$

« collvparmu_general_operation_Maxwellian_unshifted_loc »

Step 2: computation of α coefficients $f_a(\boldsymbol{r}, v_\parallel, u, t) = \sum_l \alpha_{\ell,a}(\boldsymbol{r}, v_\parallel, t) P_\ell(u)$

« collvparmu_general_alphaprojection »

Step 3: evolution of alpha coeffcients $\frac{\partial \alpha_{\ell,a}}{\partial t} = \sum_j \left[\alpha'_{i,a} N^{jl}_{0,a} + \frac{\partial \alpha'_{i,a}}{\partial v_\parallel} N^{jl}_{1,a} + \frac{\partial^2 \alpha'_{i,a}}{\partial v_\parallel^2} N^{jl}_{2,a} \right]$

« collvparmu_general_CvCd »

Step 4: reconstruction of the normalised distribution function

« collvparmu_general_reconstruction » $f_a(\boldsymbol{r}, v_\parallel, u, t) = \sum_l \alpha_{\ell,a}(\boldsymbol{r}, v_\parallel, t) P_\ell(u)$

Step 5: $f_a \longrightarrow F_a$

« collvparmu_general_operation_Maxwellian_unshifted_loc »

Step 3: evolution of alpha coeffcients

$$\frac{\partial \boldsymbol{\alpha}}{\partial \hat{t}} = T\boldsymbol{\alpha} + \boldsymbol{S}$$

Due to $f_a \neq g_a$

$$\boldsymbol{\alpha} = \begin{pmatrix} \vdots \\ \alpha_0^{(k)} \\ \alpha_1^{(k)} \\ \vdots \\ \alpha_{Npol-1}^{(k)} \\ \alpha_0^{(k+1)} \\ \vdots \end{pmatrix}$$

$$\boldsymbol{S} = \begin{pmatrix} \vdots \\ S_0^{(k)} = \kappa_{0,a}^{(k)} \hat{N}_{0,a}^{jl} + \frac{\partial \kappa_{0,a}^{(k)}}{\partial \hat{v}_\parallel} \hat{N}_{1,a}^{jl} + \frac{\partial^2 \kappa_{0,a}^{(k)}}{\partial \hat{v}_\parallel^2} \hat{N}_{2,a}^{jl} \\ S_1^{(k)} = \kappa_{1,a}^{(k)} \hat{N}_{1,a}^{jl} + \frac{\partial \kappa_{1,a}^{(k)}}{\partial \hat{v}_\parallel} \hat{N}_{1,a}^{jl} + \frac{\partial^2 \kappa_{1,a}^{(k)}}{\partial \hat{v}_\parallel^2} \hat{N}_{2,a}^{jl} \\ S_2^{(k)} = 0 \\ \vdots \\ S_{Npol-1}^{(k)} = 0 \\ \vdots \end{pmatrix}$$

Dense matrix of size 3x3 parametrized by v|| (index k)

$$T = \begin{pmatrix} B_0 & C_0 & 0 & 0 & 0 & 0 & 0 \\ A_1 & B_1 & C_1 & 0 & 0 & 0 & 0 \\ 0 & A_2 & B_2 & C_2 & 0 & 0 & 0 \\ 0 & 0 & . & . & . & 0 & 0 \\ 0 & 0 & 0 & . & . & . & 0 \\ 0 & 0 & 0 & 0 & . & . & C_{k_{max}-1} \\ 0 & 0 & 0 & 0 & 0 & A_{k_{max}} & B_{k_{max}} \end{pmatrix}$$

$N_{v||}$ blocks (~128/256)

$$\begin{cases} a_{lj}^{(k)} = -\frac{\hat{N}_{1,a}^{jl(k)}}{2\Delta v_\parallel} + \frac{\hat{N}_{2,a}^{jl(k)}}{\Delta v_\parallel^2} \\ b_{lj}^{(k)} = \hat{N}_{0,a}^{jl(k)} - 2\frac{\hat{N}_{2,a}^{jl(k)}}{\Delta v_\parallel^2} \\ c_{lj}^{(k)} = \frac{\hat{N}_{1,a}^{jl(k)}}{2\Delta v_\parallel} + \frac{\hat{N}_{2,a}^{jl(k)}}{\Delta v_\parallel^2} \end{cases}$$

Step 3: evolution of alpha coefficients

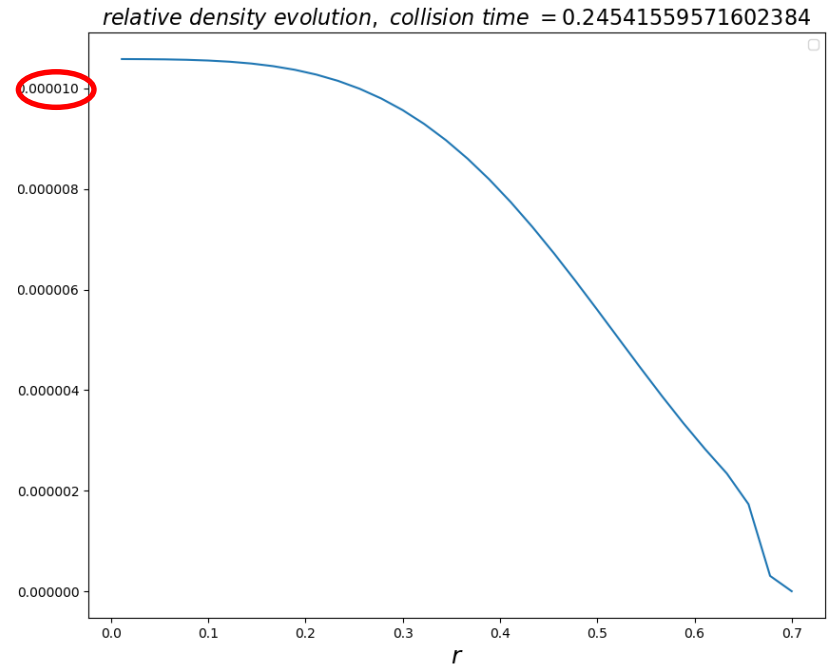
$$\frac{\partial \boldsymbol{\alpha}}{\partial \hat{t}} = T\boldsymbol{\alpha} + \boldsymbol{S}$$

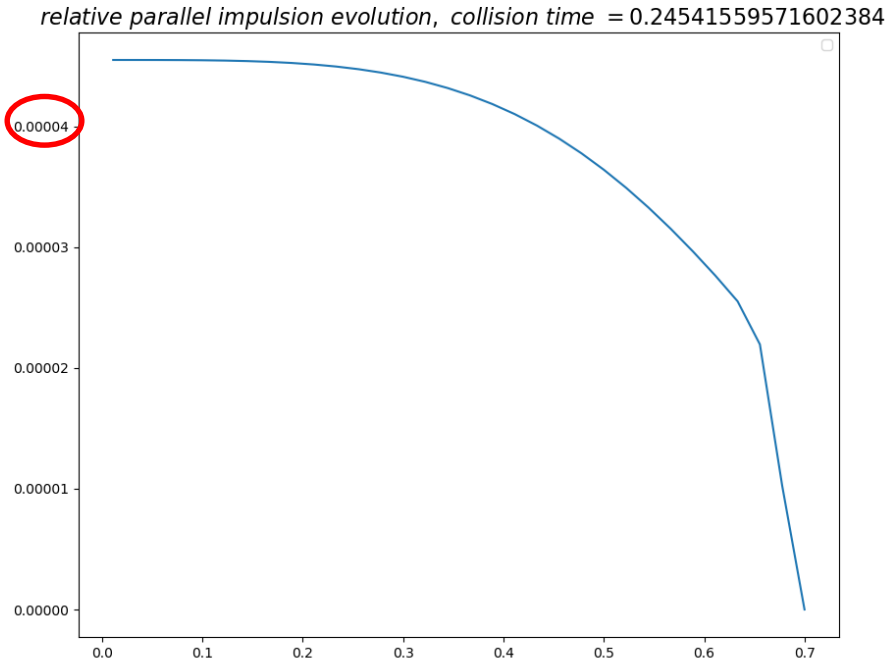
tridiagonal by blocks (blockwise generalized diagonally dominant matrices if $v\Delta t \ll 1$)

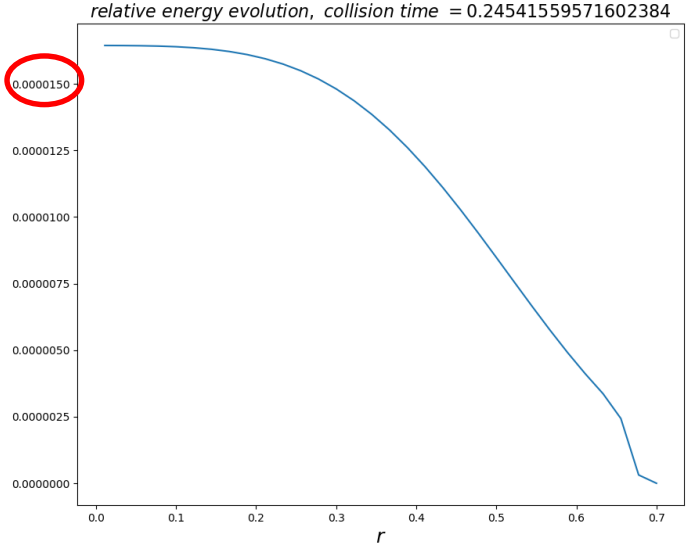
$$\begin{cases} \left(I - \frac{\Delta t}{4} T^n\right) \tilde{\boldsymbol{\alpha}} & = \left(I + \frac{\Delta t}{4} T^n\right) \boldsymbol{\alpha}^n \\ \tilde{\tilde{\boldsymbol{\alpha}}} & = \tilde{\boldsymbol{\alpha}} + \Delta t \boldsymbol{S}^n \\ \left(I - \frac{\Delta t}{4} T^n\right) \boldsymbol{\alpha}^{n+1} & = \left(I + \frac{\Delta t}{4} T^n\right) \tilde{\tilde{\boldsymbol{\alpha}}} \end{cases}$$

Solved with LU factorization

Numerical tests: conservation properties





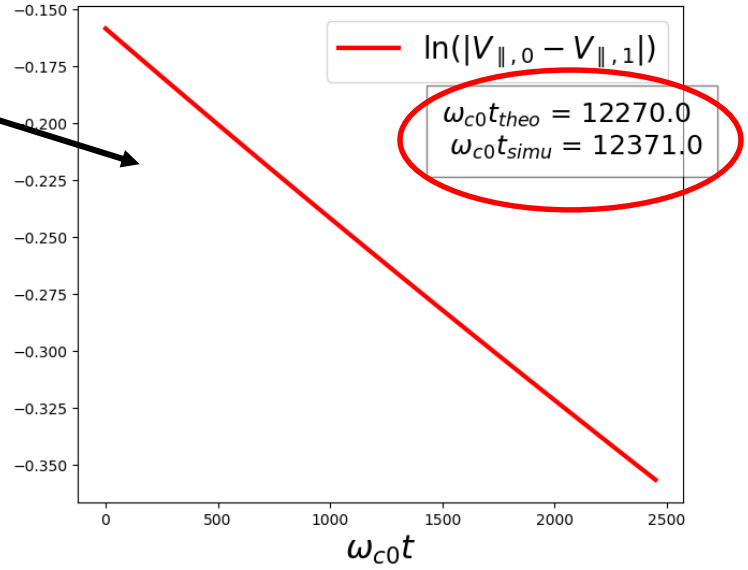


Nom de la simulation:
test_collisions_1species_conservation

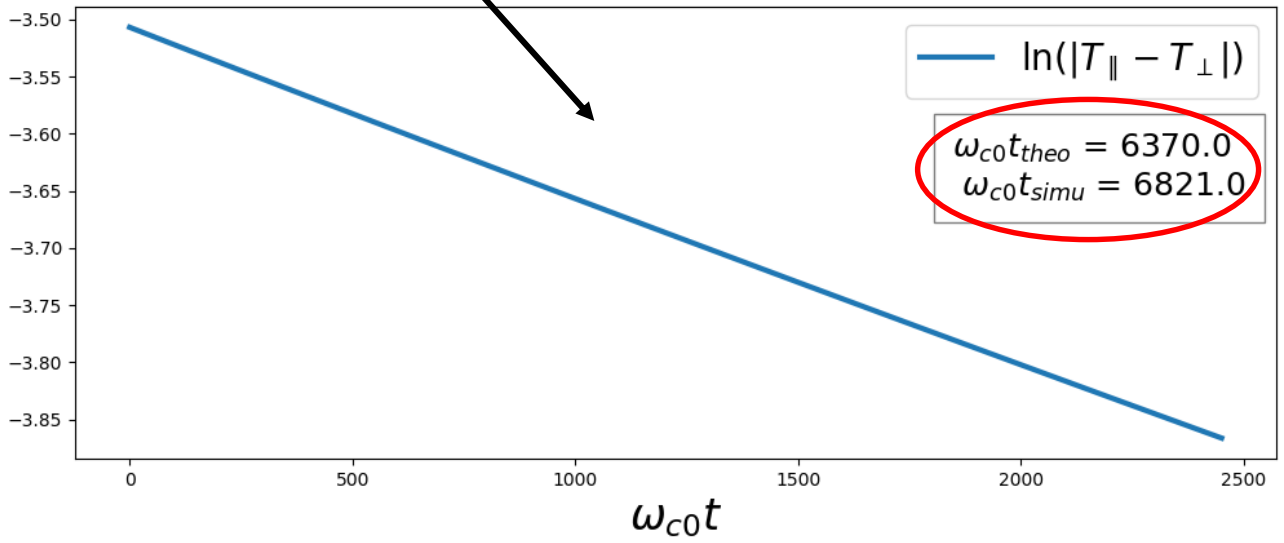
Conservations de
l'ordre de 10^-5

Numerical tests: dynamics

Nom de la simulation:
test_collisions_2species_different_V



Nom de la simulation:
test_collisions_1species_isotropisation



Nom de la simulation:
test_collisions_2species_different_T

